

UNIT II | Morphology — External Structure of Plants (12 Lectures)

1.1 Basic Concepts — Thallus vs Plant Body

Feature	Thallus	Plant Body (Cormus)
Differentiation	Absent — body undivided	Root, Stem, Leaf distinct
Vascular tissue	Absent	Present (xylem + phloem)
Examples	Spirogyra, Ulva, Marchantia	Ferns, Gymnosperms, Angiosperms
Level	Lower plants (algae, fungi)	Higher plants

1.2 Modifications of Roots

Root is the **underground, non-green, positively geotropic** part developing from the radicle. It anchors the plant and absorbs water + minerals. Has no nodes, internodes or leaves. Root tip is protected by the **root cap**; absorption occurs through **root hairs**.

Modification	Type	Example
Fusiform	Food storage (tap root)	Radish (<i>Raphanus sativus</i>)
Napiform	Food storage (tap root)	Turnip (<i>Brassica rapa</i>)
Conical	Food storage (tap root)	Carrot (<i>Daucus carota</i>)
Tuberous	Food storage (adventitious)	Sweet potato
Prop roots	Mechanical support	Banyan (<i>Ficus benghalensis</i>)
Stilt roots	Mechanical support	Maize, Sugarcane
Pneumatophores	Respiration (breathing)	Rhizophora, Avicennia (mangroves)
Nodulated roots	N ₂ fixation	Pea, Bean, Groundnut (<i>Rhizobium</i>)
Haustorial roots	Parasitism	Cuscuta (Amarbel)
Epiphytic roots	Moisture absorption from air	Vanda (orchid)

1.3 Modifications of Stem

Type	Category	Description	Example
Rhizome	Underground	Horizontal; nodes & internodes present	Ginger, Turmeric, Lotus
Tuber	Underground	Swollen tip; "eyes" = buds	Potato
Bulb	Underground	Condensed stem; fleshy scale leaves	Onion, Garlic
Corm	Underground	Solid, vertical, swollen stem	Colocasia, Gladiolus

Runner	Sub-aerial	Creeps on soil; roots at nodes	Grass, Strawberry
Stolon	Sub-aerial	Arches and touches ground	Jasmine, Mint
Tendrill	Aerial	Slender; helps climb	Cucumber, Passion flower
Thorn	Aerial	Hard, sharp; protection	Citrus, Bougainvillea
Phylloclade	Aerial	Flattened green stem; photosynthesis	Cactus (Opuntia)

1.4 Modifications of Leaf

Modification	Purpose	Example
Spines	Protection from herbivores	Cactus (leaves → spines), Barberry
Leaf tendrils	Climbing support	Pea (<i>Pisum sativum</i>) — leaflets become tendrils
Fleshy leaves	Water / food storage	Onion scale leaves, Aloe vera
Pitcher (<i>Nepenthes</i>)	Insect trapping (N-poor soil)	Pitcher plant
Phyllode	Photosynthesis (petiole flattened)	Australian <i>Acacia</i> spp.
Foliar buds	Vegetative reproduction	<i>Bryophyllum</i> — plantlets at leaf margins

1.5 Flower, Inflorescence, Fruits and Seeds

A **flower** is the reproductive structure of angiosperms consisting of four whorls: Calyx (sepals), Corolla (petals), Androecium (stamens — male) and Gynoecium (pistil — female).

Fruit Type	Characteristics	Examples
Drupe	Fleshy; hard stone (endocarp)	Mango, Coconut, Plum
Berry	Fleshy; no hard inner layer	Tomato, Grape, Banana
Legume/Pod	Dry; dehiscent along two sutures	Pea, Bean, Mustard
Capsule	Dry; many seeds; various dehiscence	Cotton, Poppy, <i>Datura</i>
Aggregate	Many free carpels of one flower	Raspberry, Strawberry
Multiple/Composite	Entire inflorescence	Jackfruit, Pineapple, Mulberry